

H.S. 2nd Year — Computer Science and Applications (2023)

Solved Question Paper — Code: 23E—CSCA (Bengali)

Full Marks: 70 | Pass Marks: 21 | Time: Three hours

Q. No. 1 — Answer any eight ($1 \times 8 = 8$) [All parts solved]

(a) How many basic number systems are there ?

Ans: There are four basic number systems commonly used in computers: (1) Binary (Base 2), (2) Octal (Base 8), (3) Decimal (Base 10), and (4) Hexadecimal (Base 16).

(b) What is a compiler ?

Ans: A compiler is a system software that translates an entire high-level language source program (e.g., a C++ program) into machine/object code (or another language) all at once before execution, and reports all errors together after scanning the whole program.

(c) What is 'case selection' in programming language ?

Ans: Case selection refers to a control statement (the switch-case statement) used in programming languages to select and execute one of many possible blocks of code, based on the value of a given expression/variable, as an alternative to writing multiple if-else statements.

(d) What is GUI ?

Ans: GUI stands for Graphical User Interface. It is a visual way of interacting with a computer using graphical elements like icons, buttons, menus, and windows (instead of typing text commands), operated using a mouse, touchscreen, or keyboard.

(e) What is a variable in programming ?

Ans: A variable is a named storage location in a computer's memory that is used to hold a data value, which can be changed/modified during the execution of a program. Each variable has a data type (e.g., int, float, char) that defines what kind of value it can store.

(f) What is a 'header file' ?

Ans: A header file is a file (with extension .h, or without extension in modern C++, e.g., <iostream>) that contains declarations of built-in functions, classes, and constants, which need to be included in a program using the #include directive before those functions/classes can be used, e.g., <iostream>, <math.h>.

(g) What is the full form of ASCII ?

Ans: The full form of ASCII is American Standard Code for Information Interchange. It is a character-encoding standard used to represent text (letters, digits, symbols) as numeric codes in computers.

(h) What is a 'keyword' ?

Ans: A keyword (or reserved word) is a word that has a special, predefined meaning in a programming language and is reserved for a specific purpose by the compiler, so it cannot be used as an identifier (variable/function name), e.g., int, if, else, for, class, return in C++.

(i) What do you mean by 'Read Only Memory' ?

Ans: Read Only Memory (ROM) is a type of non-volatile primary memory in a computer whose content can only be read, not normally modified/written by the user during program execution. It retains its data even when the power is switched off, and is typically used to store permanent/firmware instructions like the BIOS.

(j) What is RISC ?

Ans: RISC stands for Reduced Instruction Set Computer. It is a type of computer processor architecture that uses a small, simplified set of instructions, each of which can be executed in a single clock cycle, aiming for faster and more efficient performance compared to CISC (Complex Instruction Set Computer).

Q. No. 2 — Answer any ten ($2 \times 10 = 20$) [All parts solved]

(a) What are the functions of 'cout' and 'cin' ?

Ans: cout (character output) is a predefined object in C++ (defined in `<iostream>`), used along with the insertion operator (`<<`) to display/output data on the standard output device (screen), e.g., `cout << "Hello";` cin (character input) is a predefined object used along with the extraction operator (`>>`) to read/accept input data from the standard input device (keyboard), e.g., `cin >> x;`

(b) What is the difference between 'class' and 'structure' ?

Ans: Class: (1) By default, all members (data members and member functions) of a class are private. (2) It is generally used when data hiding/encapsulation is the primary requirement, along with OOP features. Structure: (1) By default, all members of a structure are public. (2) It is generally used to group together simple related data items without emphasising data hiding. (In C++, apart from this default access specifier difference, class and struct are functionally almost identical.)

(c) What is the difference between 'break' and 'continue' ?

Ans: break: (1) It is used to immediately terminate/exit the entire loop (or switch statement) in which it is used, and control passes to the statement immediately after the loop. continue: (1) It is used to skip the remaining statements of the current iteration of the loop and move directly to the next iteration (re-checking the loop condition), without terminating the loop altogether.

(d) Mention the basic features of OOP.

Ans: The basic features of Object-Oriented Programming (OOP) are: (1) Class and Object, (2) Data Encapsulation (binding data and functions together), (3) Data Abstraction (hiding implementation details, showing only essential features), (4) Inheritance (acquiring properties of another class), (5) Polymorphism (one interface, many forms), and (6) Data Hiding (restricting direct access to data).

(e) Write the main characteristics of a constructor.

Ans: Main characteristics of a constructor: (1) It has the same name as the class in which it is declared. (2) It does not have any return type, not even void. (3) It is automatically/implicitly invoked whenever an object of the class is created. (4) It can be overloaded — a class can have more than one constructor with different parameter lists. (5) It is mainly used to initialise the data members of a class.

(f) What is the difference between ++i and i++ ?

Ans: ++i (pre-increment): The value of i is incremented first, and then the updated (new) value is used in the expression. i++ (post-increment): The current (old) value of i is used first in the expression, and then i is incremented afterward. For example, if i = 5: in 'x = ++i;' both i and x become 6; but in 'x = i++;' x becomes 5, and then i becomes 6.

(g) What is 'enum' ? Give one example.

Ans: enum (enumeration) is a user-defined data type in C++ that consists of a set of named integer constants, making a program more readable by allowing the use of meaningful names instead of numbers. Example: `enum Day {Sun, Mon, Tue, Wed, Thu, Fri, Sat};` — here Sun = 0, Mon = 1, and so on, by default.

(h) Write about the data hiding in OOP.

Ans: Data hiding is an OOP concept in which the internal data (data members) of a class are hidden from direct access by outside code/other classes, by declaring them as 'private' (or 'protected'). The data can only be accessed or modified indirectly, through the public member functions of the class. This protects data from accidental corruption or unauthorised access, and is a core part of the encapsulation feature of OOP.

(i) What is the use of multi-dimensional array ?

Ans: A multi-dimensional array is used to store data in the form of a table or grid (i.e., data organised in rows and columns, or more dimensions), which is useful for representing matrices, tables, and other multi-index data — e.g., a two-dimensional array `int a[3][4]`; can be used to store and process a 3×4 matrix, or marks of 3 students in 4 subjects.

(j) What is logical error ?

Ans: A logical error is a type of error in a program in which the code compiles and runs successfully without any error message, but produces incorrect or unexpected output/results, because the underlying logic/algorithm used to solve the problem is flawed, e.g., using '+' instead of '*' in a formula.

(k) What is the use of utility software ?

Ans: Utility software is a type of system software designed to help analyse, configure, optimise, and maintain a computer system, so that it runs efficiently. Uses include: virus scanning (antivirus), disk defragmentation, file compression/backup, disk clean-up, and file management.

(l) State the types of 'VIRUS'.

Ans: Common types of computer viruses: (1) Boot sector virus – infects the boot sector of a storage device. (2) File infector virus – attaches itself to executable program files. (3) Macro virus – infects documents that use macros (e.g., Word/Excel files). (4) Polymorphic virus – changes its code/signature each time it infects a new file, to avoid detection. (5) Resident virus – installs itself in the computer's memory and infects files as they are opened.

Q. No. 3 — Answer any nine ($3 \times 9 = 27$) [All parts solved]

(a) If base and power are given, write a program in C++ to calculate the result.

```
#include <iostream>
using namespace std;
int main()
{
    int base, power, i;
    long result = 1;
    cout << "Enter base: "; cin >> base;
    cout << "Enter power: "; cin >> power;
    for (i = 1; i <= power; i++)
        result = result * base;
    cout << base << " raised to power " << power << " = " << result << endl;
    return 0;
}
```

(b) Write the difference between Object-oriented programming and Structured programming.

Ans: Structured (procedural) programming: (1) The program is divided into small functions/procedures, and importance is given to functions rather than data. (2) Data moves openly between functions and is generally not protected/secured (no data hiding). (3) Example languages: C, Pascal, FORTRAN. Object-oriented programming: (1) The program is divided into objects, which combine data and the functions that operate on that data, and importance is given to data. (2) Data is protected using data hiding/encapsulation and can only be accessed through member functions. (3) Example languages: C++, Java, Python.

(c) Write a C++ program using multi-dimensional array for matrix addition.

```
#include <iostream>
using namespace std;
int main()
{
    int a[3][3], b[3][3], c[3][3], i, j;
    cout << "Enter elements of first 3x3 matrix:\n";
```

```

    for (i = 0; i < 3; i++)
        for (j = 0; j < 3; j++)
            cin >> a[i][j];
    cout << "Enter elements of second 3x3 matrix:\n";
    for (i = 0; i < 3; i++)
        for (j = 0; j < 3; j++)
            cin >> b[i][j];
    for (i = 0; i < 3; i++)
        for (j = 0; j < 3; j++)
            c[i][j] = a[i][j] + b[i][j];
    cout << "\nSum of the two matrices:\n";
    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
            cout << c[i][j] << "\t";
        cout << endl;
    }
    return 0;
}

```

(d) Write about different operators available in C++.

Ans: C++ provides several categories of operators: (1) Arithmetic operators – +, -, *, /, % (used for basic mathematical operations). (2) Relational operators – ==, !=, <, >, <=, >= (used to compare two values). (3) Logical operators – && (AND), || (OR), ! (NOT) (used to combine conditions). (4) Assignment operators – =, +=, -=, *=, /= (used to assign values to variables). (5) Increment/Decrement operators – ++, -- (used to increase/decrease a value by 1). (6) Bitwise operators – &, |, ^, ~, <<, >> (operate on individual bits). (7) Conditional (ternary) operator – ?: (a shorthand for if-else). (8) Special operators – sizeof, scope resolution (::), new, delete.

(e) Write about the benefits of GUI-based operating system.

Ans: Benefits of a GUI-based operating system: (1) Easy to use – users interact through visual icons/menus rather than remembering and typing text commands. (2) User-friendly for beginners, reducing the learning curve. (3) Supports multitasking conveniently through multiple windows. (4) Visually represents files/folders, making file management easier (drag-and-drop). (5) Reduces the chances of typing errors, as most actions are performed using a mouse/touch instead of the keyboard.

(f) Write a C++ program to find the length of a string.

```

#include <iostream>
#include <cstring>
using namespace std;
int main()
{
    char str[100];
    cout << "Enter a string: ";
    cin.getline(str, 100);
    int length = strlen(str);
    cout << "Length of the string = " << length << endl;
    return 0;
}

```

Sample Output:

```

Enter a string: Karimganj College
Length of the string = 18

```

(g) Write a C++ program to find the total of three numbers.

```

#include <iostream>

```

```
using namespace std;
int main()
{
    float a, b, c, total;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    total = a + b + c;
    cout << "Total = " << total << endl;
    return 0;
}
```

(h) Write a C++ program to make a pyramid of numbers.

```
#include <iostream>
using namespace std;
int main()
{
    int n, i, j;
    cout << "Enter number of rows: "; cin >> n;
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= i; j++)
            cout << j << " ";
        cout << endl;
    }
    return 0;
}
```

Sample Output (for n = 4):

```
1
1 2
1 2 3
1 2 3 4
```

(i) Write about the different access modifiers in C++.

Ans: C++ has three access modifiers (access specifiers): (1) Private – members declared private are accessible only within the same class; not accessible outside the class or by derived classes (this is the default for a class). (2) Public – members declared public are accessible from anywhere the object is visible, including outside the class. (3) Protected – members declared protected are accessible within the same class and within its derived (child) classes, but not from outside the class hierarchy.

(j) Write a C++ program to store information about a person using structure.

```
#include <iostream>
using namespace std;
struct Person
{
    char name[30];
    int age;
    char address[50];
};
int main()
{
    Person p;
    cout << "Enter name: "; cin >> p.name;
    cout << "Enter age: "; cin >> p.age;
    cout << "Enter address: "; cin >> p.address;
```

```

    cout << "\nPerson Details:\n";
    cout << "Name      : " << p.name << endl;
    cout << "Age       : " << p.age << endl;
    cout << "Address   : " << p.address << endl;
    return 0;
}

```

(k) Write a C++ program to use 'switch-case'.

```

#include <iostream>
using namespace std;
int main()
{
    int day;
    cout << "Enter day number (1-7): "; cin >> day;
    switch (day)
    {
        case 1: cout << "Sunday"; break;
        case 2: cout << "Monday"; break;
        case 3: cout << "Tuesday"; break;
        case 4: cout << "Wednesday"; break;
        case 5: cout << "Thursday"; break;
        case 6: cout << "Friday"; break;
        case 7: cout << "Saturday"; break;
        default: cout << "Invalid day number";
    }
    return 0;
}

```

Q. No. 4 — Answer any three ($5 \times 3 = 15$) [All parts solved]

(a) Write a program in C++ to take the marks of a student and prepare the result sheet.

```

#include <iostream>
using namespace std;
int main()
{
    char name[30];
    int eng, math, sci, comp, social, total;
    float percentage;
    cout << "Enter student name: "; cin >> name;
    cout << "Enter marks in English, Maths, Science, Computer, Social Science:\n";
    cin >> eng >> math >> sci >> comp >> social;
    total = eng + math + sci + comp + social;
    percentage = (float)total / 5;
    cout << "\n----- RESULT SHEET ----- \n";
    cout << "Name      : " << name << endl;
    cout << "English   : " << eng << endl;
    cout << "Maths     : " << math << endl;
    cout << "Science   : " << sci << endl;
    cout << "Computer  : " << comp << endl;
    cout << "Social Sc. : " << social << endl;
    cout << "Total     : " << total << " / 500" << endl;
    cout << "Percentage : " << percentage << " %" << endl;
    if (percentage >= 33)
        cout << "Result      : PASS" << endl;
}

```

```

        else
            cout << "Result      : FAIL" << endl;
        return 0;
    }

```

(b) Write a program in C++ to determine if a number is prime.

```

#include <iostream>
using namespace std;
int main()
{
    int n, i;
    bool isPrime = true;
    cout << "Enter a number: "; cin >> n;
    if (n <= 1)
        isPrime = false;
    for (i = 2; i <= n / 2; i++)
    {
        if (n % i == 0)
        {
            isPrime = false;
            break;
        }
    }
    if (isPrime)
        cout << n << " is a Prime number." << endl;
    else
        cout << n << " is NOT a Prime number." << endl;
    return 0;
}

```

(c) Write a program in C++ to compare two strings for equality.

```

#include <iostream>
#include <cstring>
using namespace std;
int main()
{
    char str1[50], str2[50];
    cout << "Enter first string: ";
    cin.getline(str1, 50);
    cout << "Enter second string: ";
    cin.getline(str2, 50);
    if (strcmp(str1, str2) == 0)
        cout << "Both strings are EQUAL." << endl;
    else
        cout << "Strings are NOT equal." << endl;
    return 0;
}

```

(d) Write a program in C++ to reverse a number.

```

#include <iostream>
using namespace std;
int main()
{
    int n, remainder, reverse = 0;
    cout << "Enter a number: "; cin >> n;

```

```
while (n != 0)
{
    remainder = n % 10;
    reverse = reverse * 10 + remainder;
    n = n / 10;
}
cout << "Reversed number = " << reverse << endl;
return 0;
}
```

Sample Output:

Enter a number: 12345
Reversed number = 54321